

Introduction

Underdeveloped and rural regions often lack access to quality microscopes, even though microscopy is essential for diagnosing blood-borne diseases and overall health conditions. It is especially critical for Complete Blood Counts (CBCs), which detect conditions like anemia, infections, blood cancers and more (Cleveland Clinic, 2024). Since labs are scarce, samples must travel long distances, delaying diagnosis and treatment and allowing conditions to worsen. For example, each hour blood bacteria infection treatments are delayed increases the risk of death by 8% (McGregor, 2014).

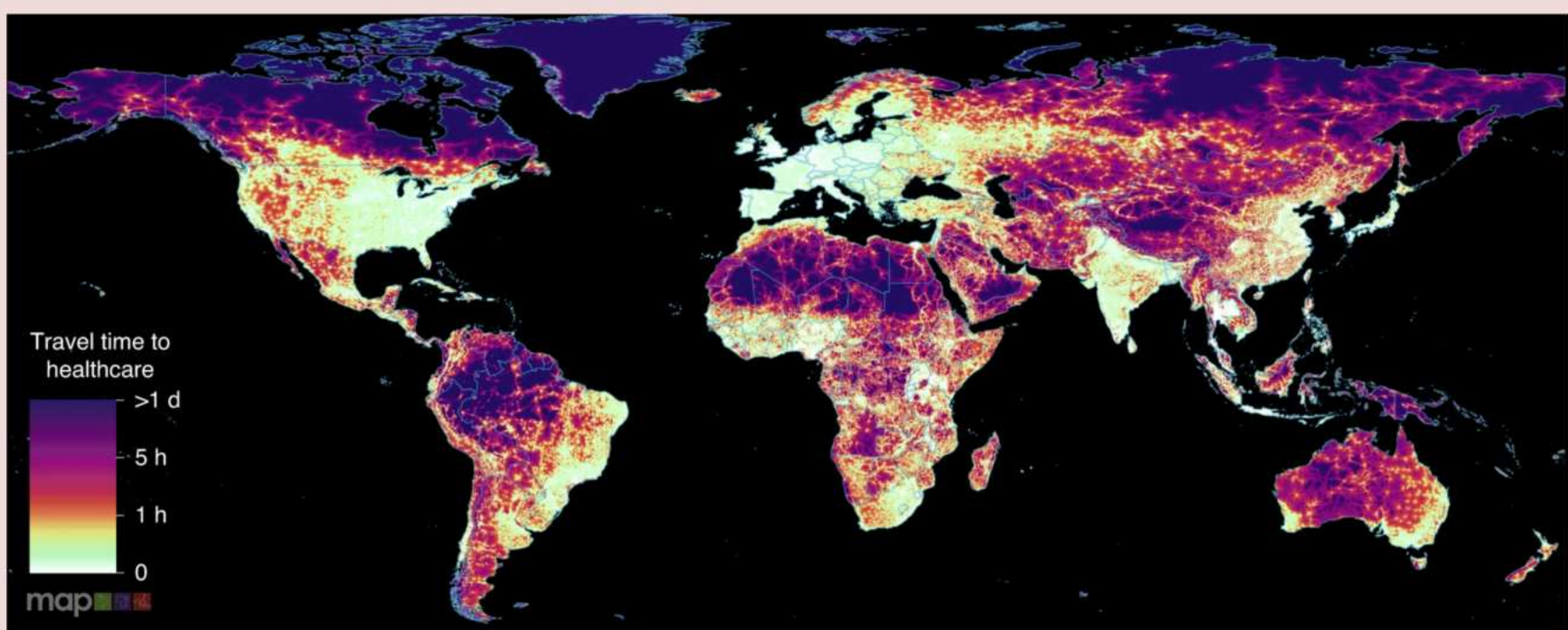


Figure 1: Map of travel time to hospitals or medical labs (Nature, 2020)

Phone microscopy has emerged as a novel way to expand access to medical diagnostics in under resourced areas, allowing patients to image blood samples from home and avoid long wait times. However, phone microscopes have many optical limitations such as low contrast, small field of view, low magnification, and aberration of features.

Purpose

This project aims to enhance phone microscope imaging so it can accurately detect and identify blood cells and provide accurate measurements of their characteristics. This improvement will enable faster diagnoses and earlier treatment for patients in under resourced communities.

Developing a Plasmonically Enhanced Phone Microscope System and Deconvolution Model to Improve Blood Microscopy Diagnosis in Underresourced Areas

Methods

Enhanced Optical System

Laser Pointer Lens:

- Increases effective magnification
- Improves light collimation into the phone microscope

Plasmonically Enhanced Gold Nanoparticle Coverslip:

- Amplifies the light reflected from cells
- Boosts weak signals for smartphone detection
- Improved signal-to-noise ratio



Figure 2: Enhanced Optical System (Picture and Prototype created by Finalist)

Deconvolution Model

- Uses iterative maximum likelihood estimation to jointly optimize the image and unknown Point Spread Function
- Refines the PSF each iteration to correct sample-specific optical distortions.
- Sharpens features by redistributing blurred intensity toward its most probable origin.

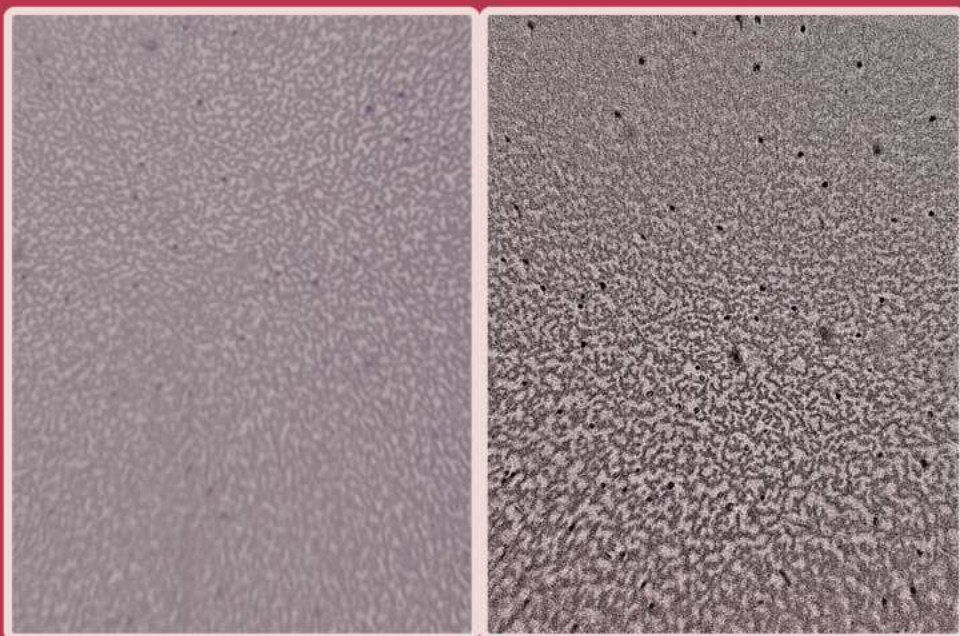


Figure 3: Blood Smear Before vs After Model (Picture taken by Finalist)

Testing

- Imports raw, enhanced, and lab-grade blood smear images into ImageJ for analysis.
- Quantifies blood cell counts, total area, and average cell size using ImageJ outputs.
- Compares raw and enhanced results to lab-grade values using percent difference to assess accuracy improvement.

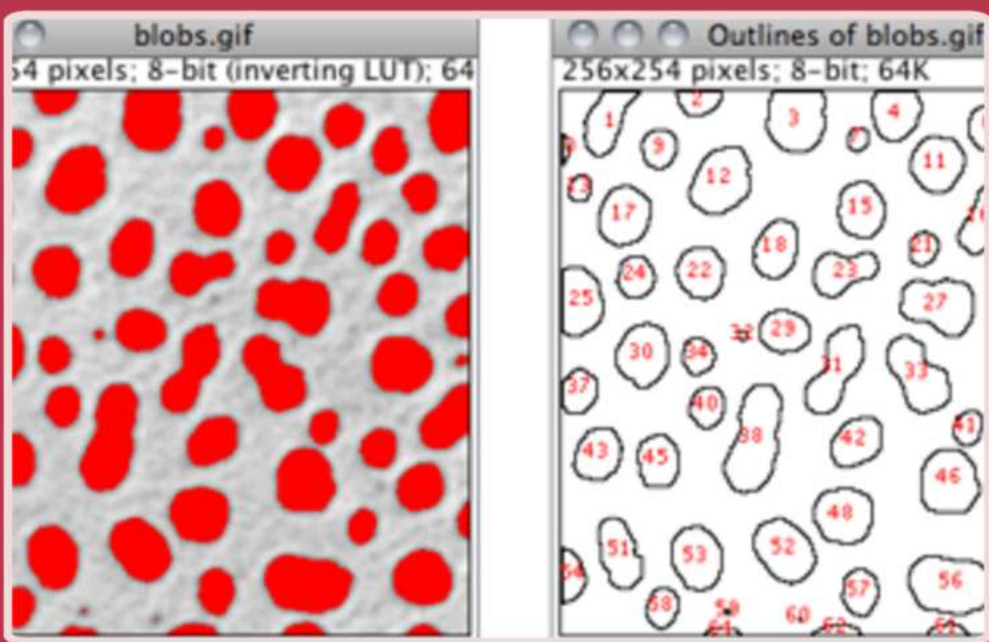


Figure 4: Image J Thresholding and Outlining Cells from an Image (Ferreira, 2017)

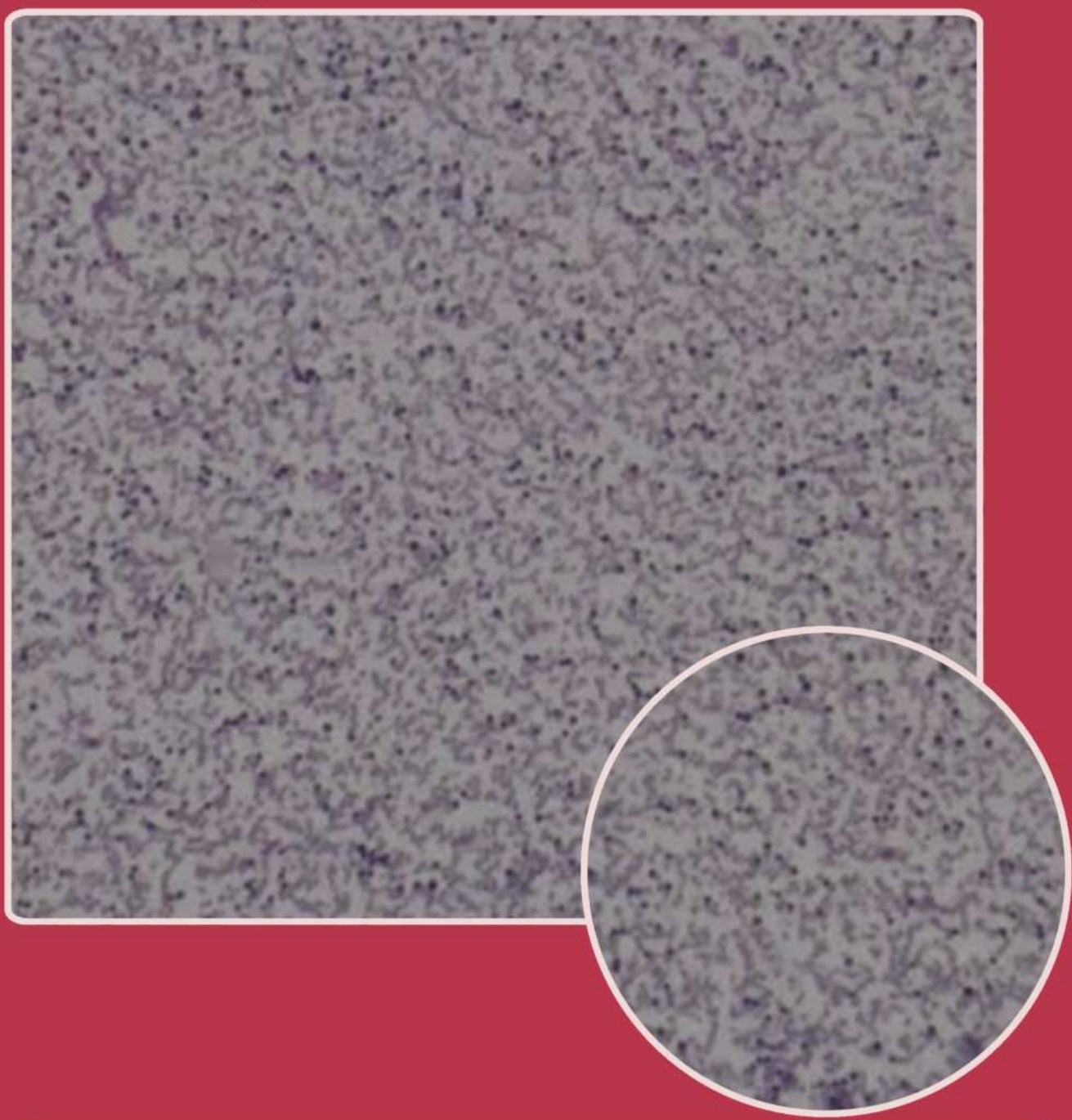


Figure 5: Original Phone Microscope Photo (Picture taken by Finalist)

Count = 5.052%
Total Area = 4.310%
Average Size = 0.993%

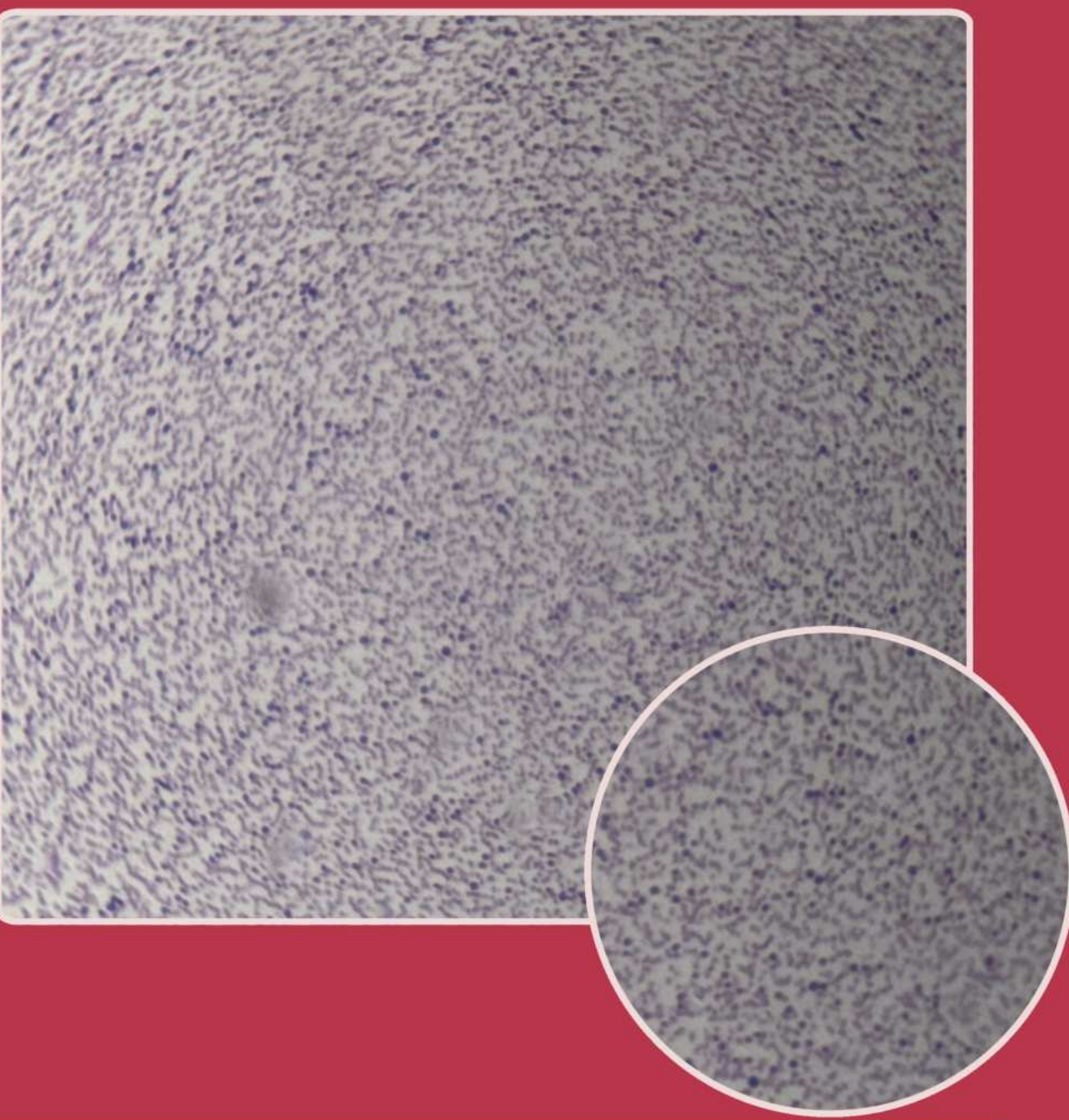


Figure 6: Enhanced Optical System (Picture taken by Finalist)

Count = 4.293%
Total Area = 3.147%
Average Size = 0.615%



Figure 7: Deconvoluted Image (Picture taken by Finalist)

Count = 2.515%
Total Area = 1.319%
Average Size = -0.457%

Results

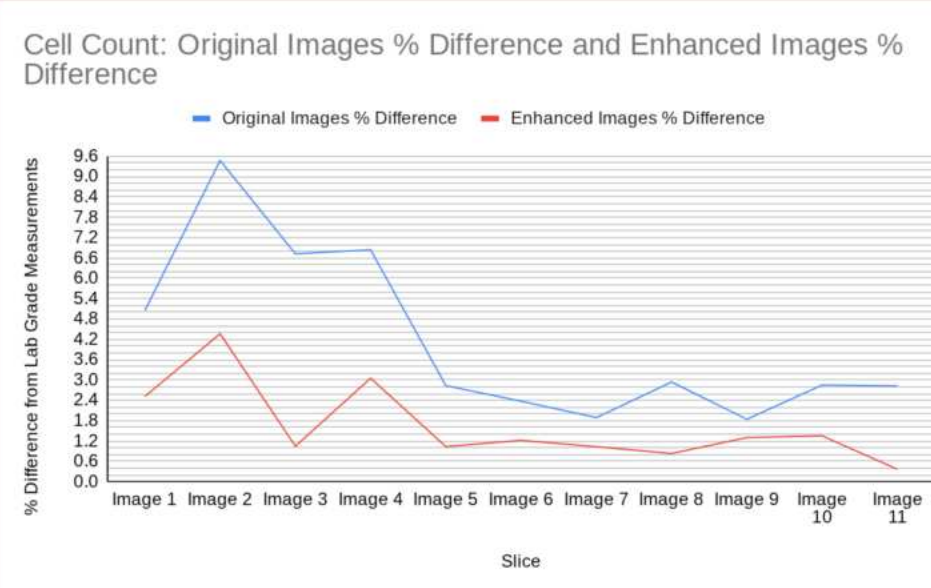


Figure 8: Graph of percent difference in cell count measurements from high quality lab grade image (Graph made by Finalist in Google Sheets)

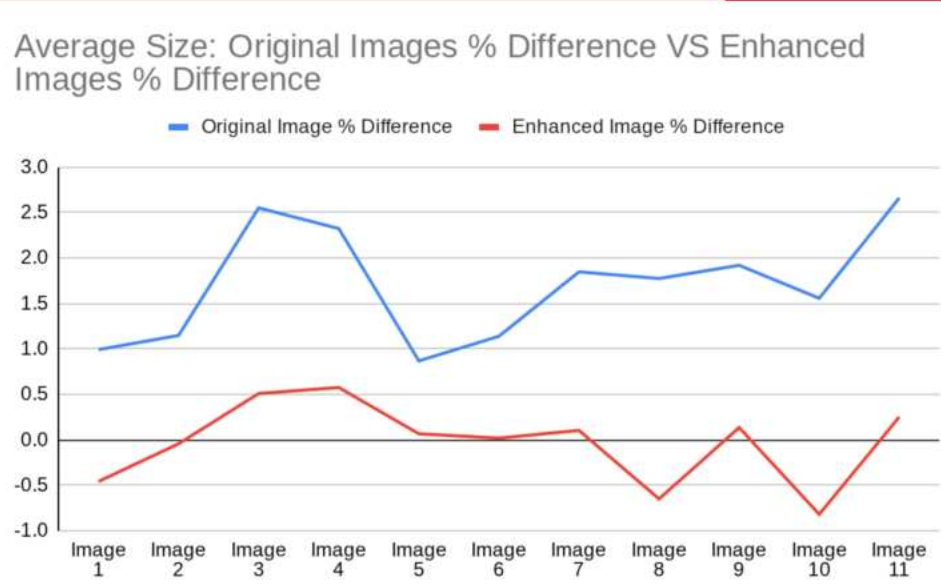


Figure 9: Graph of percent difference in average size measurements from high quality lab grade image (Graph made by Finalist in Google Sheets)

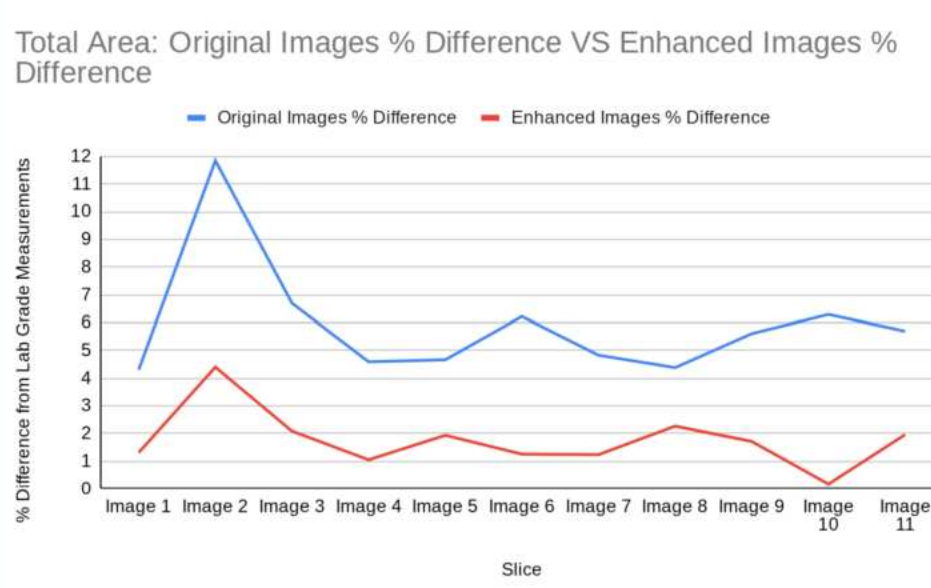


Figure 10: Graph of percent difference in total area measurements from high quality lab grade image (Graph made by Finalist in Google Sheets)

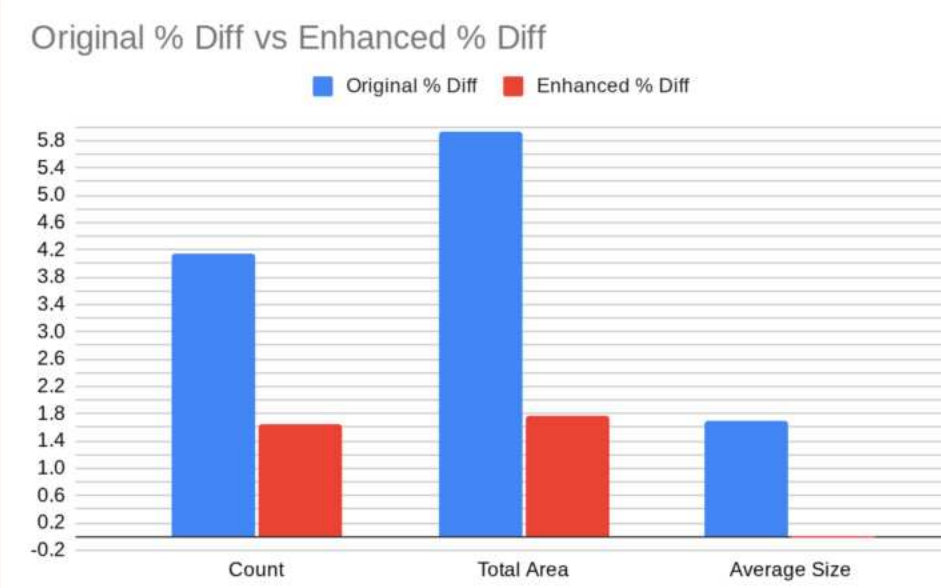


Figure 11: Bar graph of average percent difference in measurements from high quality lab grade image (Graph made by Finalist in Google Sheets)

Data shows the enhanced system that I built achieved on average 2.5% improved counting accuracy, 4.2% improved area measurement accuracy, and 1.73% improved average size measurement accuracy.

Conclusions

The results of this project show that my enhanced optical system and deconvolution model significantly improved accuracy of count, area and average size measurements. These improvements increase the diagnostic potential of phone microscopy for conditions such as infections, leukemia, and clotting disorders. Further extensions of this project include advancing color accuracy of images to expand applicability to color-dependent diagnoses such as anemia and hemolysis.

References

Cleveland Clinic. (2024, March 14). Complete blood count. Cleveland Clinic. <https://my.clevelandclinic.org/health/diagnostics/4053-complete-blood-count>
Image.sc Forum. "Displaying Major and Minor Axes on Objects." Image.sc Forum, 8 Dec. 2017. forum.image.sc/t/displaying-major-and-minor-axes-on-objects/8195. Accessed 4 Mar. 2026.
McGregor, C. (2014). Improving Time to Antibiotics and Implementing the "Sepsis 6." BMJ Quality Improvement Reports, 2(2), u202548.w1443. <https://doi.org/10.1136/bmjquality.u202548.w1443>